Week 11: Machine Consciousness: Theories and Indicator Properties

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Scientific theories of consciousness make claims about the mechanisms that give rise to experience. This week examines the global workspace, integrated information, higher-order and recurrent processing theories, and a recent method that derives indicator properties from them to assess AI systems [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to describe four major theories of consciousness and what each implies for machines. They should be able to compute a simplified integration measure for small networks. They should also be able to explain the indicator property method and its assumption of computational functionalism. Finally, they should be able to state the main disagreements about conscious AI.

Lecture

Global workspace theories

Baars proposed that consciousness corresponds to a global workspace: a limited-capacity stage on which information is broadcast to many specialised, unconscious processors [1]. Only content that wins the competition for the workspace becomes conscious and available for report and flexible use. Dehaene and Naccache developed a neuronal version in which conscious access involves a sudden, self-sustaining ignition of activity in long-range cortical networks [4]. On this family of theories, a machine with specialised modules, a limited workspace and global broadcast would satisfy the central conditions. The Colab notebook simulates ignition, a nonlinear transition from local to global activity as stimulus strength increases.

Integrated information theory

Tononi proposed that consciousness corresponds to integrated information, the extent to which a system as a whole generates information over and above its parts [2]. The theory was later formulated from axioms about the essential properties of experience, from which it derives requirements for its physical substrate [5, 6]. Its measure, Φ, is defined over the cause-effect structure of a system, and purely feedforward systems have zero Φ, whatever they compute. Integrated information theory therefore implies that a conventional digital computer running any program might have very little consciousness. The relevant quantity depends on the physical causal structure, not on the software.

The interactive lab and the Colab notebook compute a much simpler quantity: how much the whole system's past predicts its future beyond what the best split into two parts predicts. Figure 11.1 shows that this simplified measure is zero for independent or feedforward networks and positive for recurrent loops. This mirrors one qualitative claim of the theory without reproducing its formal machinery.

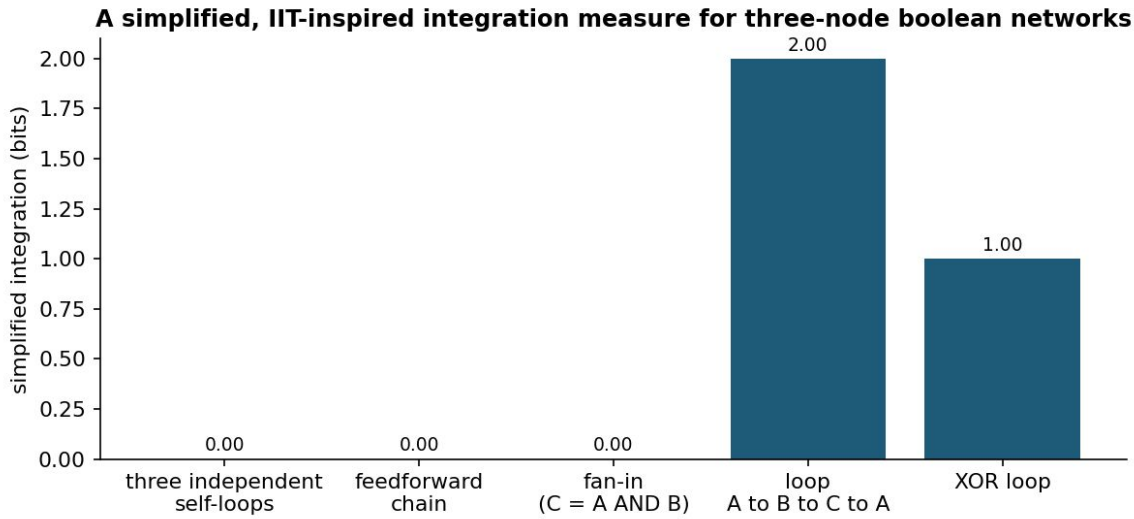


Figure 11.1. A simplified integration measure, whole-system information minus the information of the best split into parts, for five three-node Boolean networks. It is inspired by, but not equal to, Φ of integrated information theory [2, 6].

Higher-order and recurrent processing theories

Higher-order theories hold that a mental state is conscious when the subject is aware of being in it, for example through a higher-order thought about the state [7]. A machine would need to represent some of its own states as its own. Recurrent processing theory holds that consciousness arises when feedback from higher to lower visual areas produces sustained recurrent activity, even without global broadcast [8]. The theories disagree about where consciousness begins, which matters for machines because different architectures satisfy different conditions.

Indicator properties and the debate about AI

Butlin and colleagues proposed an empirical method for assessing AI systems [3]. They assumed computational functionalism as a working hypothesis and derived indicator properties from several scientific theories. The theories include recurrent processing, global workspace, higher-order, predictive processing and attention schema theories. Agency and embodiment complete the list. They assessed several recent systems and concluded that no current AI system is conscious, but also that there are no obvious barriers to building systems that satisfy the indicators. Chalmers examined large language models and argued that they lack several features that many theories require, such as recurrent processing, a global workspace and unified agency [9]. Such features could, however, be added in future systems.

Others doubt the functionalist assumption. Seth argued for biological naturalism [10]. On this view, consciousness may depend on properties of living systems, such as the regulation of a living body, so computation alone would not suffice. The interactive lab lets students weigh the theories by credence and see how the verdict about an architecture depends on those weights.

> ***Pause and reflect.*** *Would you change how you treat a system if its indicator score rose? What does your answer reveal about the role of theory in such judgements?*

Hands-on components

Interactive lab: Part A combines indicator properties with your credence in each theory to score six AI architectures. The indicator settings are illustrative starting points for discussion, not the assessments of the cited report. Part B computes a simplified integration measure for three-node networks that you design [2, 3]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-11/lab.html

Colab notebook: It implements the simplified integration measure for small Boolean networks and simulates ignition in a single global workspace unit with self-excitation [2, 4]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-11/NB11_machine_consciousness.ipynb

Reflection prompts

1. Set the credences in the interactive lab to your own beliefs. Which architecture scores highest, and would you accept that verdict?
2. The simplified measure distinguishes loops from feedforward networks. Is that difference a plausible marker of consciousness? Why or why not?
3. Which theory of this week is most testable, and what experiment would you run on an AI system to test it?

Weekly task and submission

What to do. Assess one current AI system with the indicator method. First, choose a system, such as a chat assistant based on a language model, and describe its architecture briefly. Second, examine it with at least three theories of consciousness: For each theory, state the indicator properties it implies and whether the system has them. Third, discuss whether computational functionalism, which the method assumes, is a reasonable working assumption.

Where to look. Lecture: sections "Global workspace theories", "Integrated information theory", "Higher-order and recurrent processing theories" and "Indicator properties and the debate about AI". Interactive lab: Part A for indicators and credences, and Part B for the simplified integration measure. Colab notebook: Section 1 for integration and Section 2 for ignition in a global workspace unit.

Example. Consider a robot whose controller is a recurrent network with a central planner. It has recurrence, so it meets one indicator of recurrent processing theory. Whether its planner works as a global workspace depends on whether its contents are broadcast to many modules. Integrated information theory would look at the hardware rather than at the program.

What to submit. About 600 words with the chart of your credences from Part A of the interactive lab, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Baars, B. J. (1988). A Cognitive Theory of Consciousness. Cambridge University Press.

[2] Tononi, G. (2004). An information integration theory of consciousness. BMC Neuroscience, 5, 42. https://doi.org/10.1186/1471-2202-5-42

[3] Butlin, P., Long, R., Elmoznino, E., Bengio, Y., Birch, J., Constant, A., Deane, G., Fleming, S. M., Frith, C., Ji, X., Kanai, R., Klein, C., Lindsay, G., Michel, M., Mudrik, L., Peters, M. A. K., Schwitzgebel, E., Simon, J., & VanRullen, R. (2023). Consciousness in artificial intelligence: Insights from the science of consciousness. arXiv preprint arXiv:2308.08708. https://arxiv.org/abs/2308.08708

[4] Dehaene, S., & Naccache, L. (2001). Towards a cognitive neuroscience of consciousness: Basic evidence and a workspace framework. Cognition, 79(1-2), 1-37. https://doi.org/10.1016/S0010-0277(00)00123-2

[5] Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: From consciousness to its physical substrate. Nature Reviews Neuroscience, 17(7), 450-461. https://doi.org/10.1038/nrn.2016.44

[6] Albantakis, L., Barbosa, L., Findlay, G., et al. (2023). Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. PLOS Computational Biology, 19(10), e1011465. https://doi.org/10.1371/journal.pcbi.1011465

[7] Rosenthal, D. M. (2005). Consciousness and Mind. Clarendon Press.

[8] Lamme, V. A. F. (2006). Towards a true neural stance on consciousness. Trends in Cognitive Sciences, 10(11), 494-501. https://doi.org/10.1016/j.tics.2006.09.001

[9] Chalmers, D. J. (2023). Could a large language model be conscious?. arXiv preprint arXiv:2303.07103. https://arxiv.org/abs/2303.07103

[10] Seth, A. K. (2025). Conscious artificial intelligence and biological naturalism. Behavioral and Brain Sciences. Published online 21 April 2025. https://doi.org/10.1017/S0140525X25000032